

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

6.3 Mobile IP: Learning Objectives

By the End of This Session

- Explain the Mobile IP addressing problem.
- Distinguish home and care-of addresses.
- Explain home and foreign agents.
- Describe three Mobile IP phases.
- Explain discovery and registration.
- Trace Mobile IP data transfer.
- Explain tunneling and proxy ARP.
- Analyze routing inefficiency.
- Explain the binding solution.

Concept map: Addressing → Agents → Discovery → Registration → Data Transfer

Why Mobile IP?

Motivation

- Mobile computers are increasingly popular.
- Notebooks can move between networks.
- Internet connectivity should continue.
- Mobile IP extends the IP protocol.
- Connectivity should work at different locations.

Central Problem

**What happens to an IP address when
a host moves?**

Mobile Host moving from Network A to Network B

6.3.1 Addressing: Stationary Hosts

Original IP Assumption

- Host is stationary.
- Host attaches to one network.
- Router uses IP address for routing.
- IP address has prefix and suffix.

Example

10.3.4.24/8

defines a host attached to:

10.0.0.0/8

- Prefix associates host with network.
- Suffix identifies host.

IP Address = Prefix + Suffix; Prefix identifies the attached network

Mobile Hosts: The Addressing Problem

When a Host Moves

- Host moves to another network.
- IP addressing structure must be modified.
- Original address may no longer be valid.
- Router uses prefix to locate network.

Simple Solution: Change the Address

- Use DHCP on the new network.
- Obtain a new IP address.
- Associate address with new network.

But...

Configuration, reboot, DNS updates, and transmission interruption create major drawbacks.

Changing the Address: Drawbacks

- ❶ Configuration files must change.
- ❷ Computer must be rebooted.
- ❸ DNS tables must be revised.
- ❹ Other hosts need updated information.
- ❺ Ongoing transmission may be interrupted.

Why Does Transmission Break?

- Client IP address must remain constant.
- Server IP address must remain constant.
- Port numbers must remain constant.
- Movement changes the host's IP address.

Connection: Established → Host Moves → Address Changes → Exchange Interrupted

Two Addresses: Mobile IP Solution

Home Address

- Original address.
- Permanent.
- Associated with home network.
- Identifies permanent network association.

Care-of Address

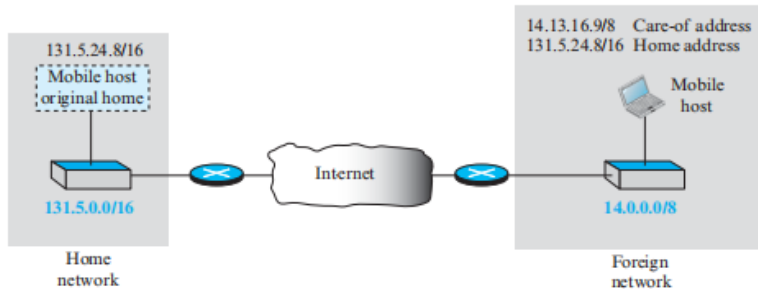
- Temporary address.
- Associated with foreign network.
- Changes after movement.
- Obtained during discovery and registration.

Core Idea

Home Address = Permanent

Care-of Address = Temporary

Figure 6.52 *Home address and care-of address*



6.3.2 Agents

Purpose

Agents make the address change transparent to the rest of the Internet.

Home Agent

- Usually a router.
- Attached to home network.
- Acts for the mobile host.
- Receives packets for mobile host.
- Sends them to foreign agent.

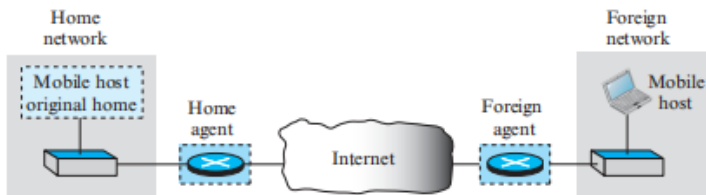
Foreign Agent

- Usually a router.
- Attached to foreign network.
- Receives home-agent packets.
- Delivers packets to mobile host.

Important

Agent function is performed at the application layer. They are both **routers and hosts**.

Figure 6.53 *Home agent and foreign agent*



Collocated Care-of Address

Mobile Host as Foreign Agent

The mobile host and foreign agent can be the same.

- Mobile host obtains care-of address itself.
- DHCP can provide the address.
- Extra software is required.
- Mobile host communicates with home agent.
- It maintains both addresses.
- Dual addressing remains transparent to applications.

Advantage

Can move to any network without depending on foreign-agent availability.

Disadvantage

Requires extra software to act as its own foreign agent.

6.3.3 Three Phases of Mobile IP

Phase 1

Agent Discovery

- Home agent
- Foreign agent
- Care-of address

Phase 2

Registration

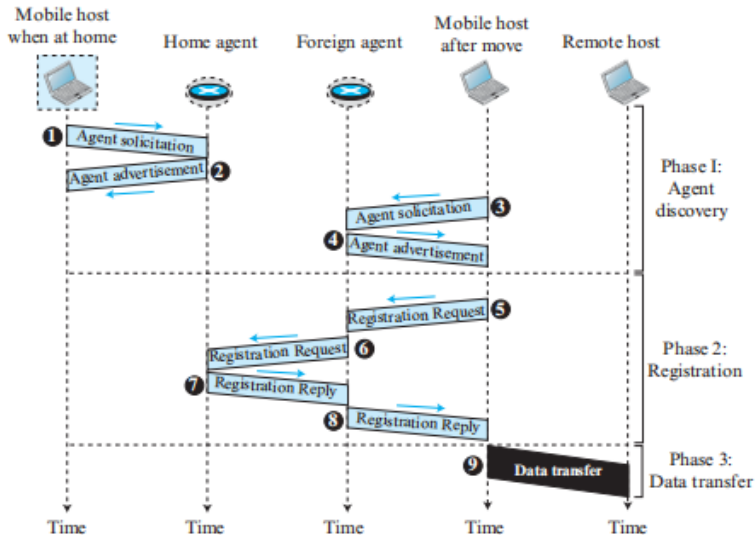
- Register
- Renew
- Deregister

Phase 3

Data Transfer

- Remote host
- Packet delivery
- Transparency

Figure 6.54 *Remote host and mobile host communication*



Phase 1: Agent Discovery

Two Subphases

- 1 Discover home agent before leaving home.
- 2 Discover foreign agent after moving.

Information Learned

- Home-agent address.
- Foreign-agent address.
- Care-of address.

Two Message Types

Advertisement

Solicitation

Home Agent Discovery → Foreign Agent Discovery

Agent Advertisement

Mechanism

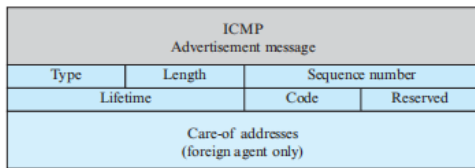
- Router advertises its presence.
- Uses an ICMP router advertisement.
- Agent can append an agent advertisement.
- Advertisement is piggybacked onto router advertisement.

Important

Mobile IP does not create a new packet type. It uses the **ICMP router advertisement packet**.

Agent Advertisement: Fields

Figure 6.55 Agent advertisement



Field	Description
Type	8-bit field; value is 16 .
Length	8-bit field; total length of extension message, not ICMP advertisement message.
Sequence Number	16-bit message number; helps determine lost messages.
Lifetime	Seconds agent accepts requests; all 1s means infinite.
Code	8-bit flag field; meanings given in Table 6.6.
Care-of Addresses	List of addresses available as care-of addresses. Mobile host chooses one. Used only by foreign agent. Selection appears in registration request.

Table 6.6: Agent Advertisement Code

Bit	Meaning
0	Registration required; no collocated care-of address.
1	Agent is busy; registration not accepted now.
2	Agent acts as a home agent.
3	Agent acts as a foreign agent.
4	Agent uses minimal encapsulation.
5	Agent uses generic routing encapsulation (GRE).
6	Agent supports header compression.
7	Unused (0).

Agent Solicitation

When Used

- Mobile host moves to a new network.
- No agent advertisement is received.
- Mobile host initiates an agent solicitation.

Mechanism

The mobile host uses an **ICMP solicitation message** to inform an agent that assistance is needed.

Important

Mobile IP uses the **ICMP router solicitation packet**.

Mobile Host → ICMP Router Solicitation → Agent

Phase 2: Registration

Four Requirements

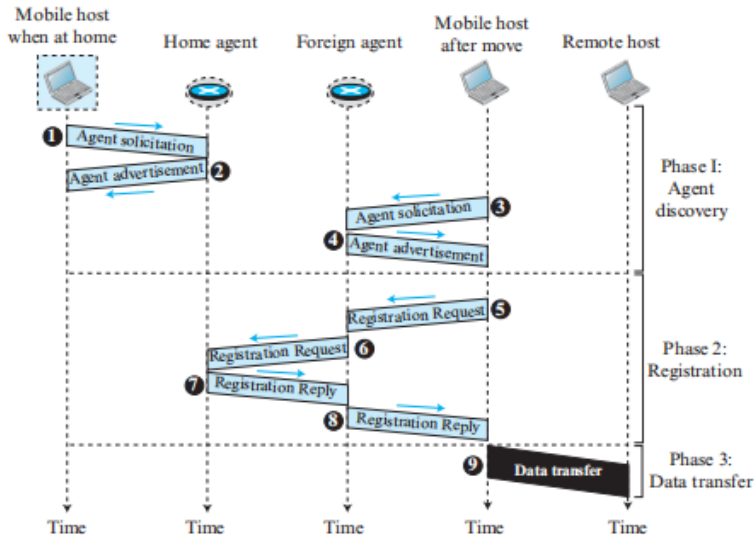
- 1 Register with the foreign agent.
- 2 Register with the home agent.
- 3 Renew registration after expiration.
- 4 Deregister when returning home.

Request and Reply

Mobile Host → Foreign Agent → Home Agent

Home Agent → Foreign Agent → Mobile Host

Figure 6.54 *Remote host and mobile host communication*



Registration Request

Purpose

The request announces:

- Care-of address.
- Home address.
- Home-agent address.

Process

- 1 Mobile host sends request.
- 2 Foreign agent receives and registers it.
- 3 Foreign agent relays request.
- 4 Home agent receives request.

Important

The home agent learns the foreign-agent address from the relaying IP packet's source address.

Figure 6.56 *Registration request format*

Type	Flag	Lifetime
Home address		
Home agent address		
Care-of address		
Identification		
Extensions ...		

Registration Request: Fields

Field	Description
Type	8-bit field; request value is 1 .
Flag	8-bit forwarding-information field; bit meanings are in Table 6.7.
Lifetime	Registration validity in seconds; all 0s = deregistration; all 1s = infinite.
Home Address	Permanent first address of mobile host.
Home Agent Address	Address of home agent.
Care-of Address	Temporary second address of mobile host.
Identification	64-bit number inserted by mobile host and repeated in reply; matches request with reply.
Extensions	Variable-length authentication extensions; allow home agent to authenticate mobile agent.

Table 6.7: Registration Request Flags

Bit	Meaning
0	Home agent retains prior care-of address.
1	Home agent tunnels any broadcast message.
2	Mobile host uses collocated care-of address.
3	Request minimal encapsulation.
4	Request generic routing encapsulation (GRE).
5	Request header compression.
6–7	Reserved bits.

Registration Reply and Encapsulation

Registration Reply

- Sent by home agent.
- Relayed through foreign agent.
- Confirms or denies request.
- Type = 3.
- Code replaces Flag.
- Code shows acceptance or denial.
- Care-of address is not needed.

Encapsulation

- Registration messages use UDP.
- Agent uses well-known port **434**.
- Mobile host uses ephemeral port.

UDP Header + Registration Message

Figure 6.57 *Registration reply format*

Type	Code	Lifetime
Home address		
Home agent address		
Identification		
Extensions ...		

Phase 3: Data Transfer

After Discovery and Registration

The mobile host can communicate with a remote host.

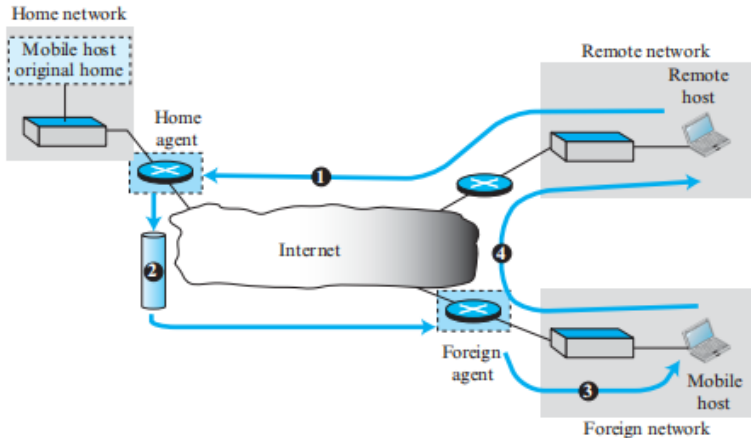
Four Data-Transfer Paths

- 1 Remote Host → Home Agent
- 2 Home Agent → Foreign Agent
- 3 Foreign Agent → Mobile Host
- 4 Mobile Host → Remote Host

Core Idea

The remote host continues using the mobile host's **home address**.

Figure 6.58 *Data transfer*



Data Transfer: Paths 1 and 2

Path 1: Remote Host → Home Agent

- Source = remote host address.
- Destination = mobile host home address.
- Home agent intercepts packet.
- Uses proxy ARP.
- Acts on behalf of mobile host.

Path 2: Home Agent → Foreign Agent

- Uses tunneling.
- Entire IP packet is encapsulated.
- Outer source = home agent.
- Outer destination = foreign agent.

Data Transfer: Paths 3 and 4

Path 3: Foreign Agent → Mobile Host

- Foreign agent removes encapsulation.
- Original destination is home address.
- Registry table is consulted.
- Care-of address is found.
- Packet is sent to care-of address.

Path 4: Mobile Host → Remote Host

- Mobile host sends normally.
- Source = home address.
- Destination = remote host.
- Packet comes from foreign network.

Figure 6.58 – Highlight Paths 3 and 4; show registry lookup and reverse transmission

Transparency in Mobile IP

Remote Host's View

The remote host is unaware of mobile-host movement.

Packets To Mobile Host

Destination:

Home Address

Packets From Mobile Host

Source:

Home Address

Result

Movement is totally transparent. The rest of the Internet is unaware of mobility.

Before/after mobility: Remote host continues using the same home address

6.3.4 Inefficiency: Double Crossing

Scenario

- Mobile host moves to remote host's network/site.
- Communication becomes geographically local.

Mobile → Remote

Local communication
No inefficiency

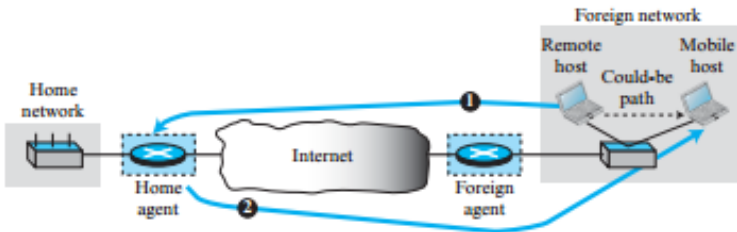
Remote → Mobile

Packet crosses
the Internet twice.

Double Crossing / 2X

Severe inefficiency. Significant because of the principle of locality.

Figure 6.59 *Double crossing*



Triangle Routing / Dog-Leg Routing

Scenario

Remote and mobile hosts are on different networks/sites.

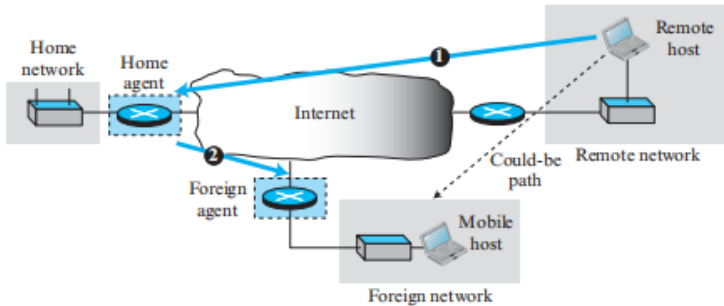
Packet Paths

- Mobile → Remote: no inefficiency.
- Remote → Mobile: Remote → Home Agent → Mobile.

Why Triangle?

Packet travels two sides of a triangle, instead of one direct side.

Figure 6.60 *Triangle routing*



Solution: Binding and Cache

Binding Solution

Bind the care-of address to the home address.

- 1 Home agent receives first packet.
- 2 Home agent forwards it to foreign agent.
- 3 Home agent sends binding update.
- 4 Remote host stores the binding.
- 5 Future packets use the care-of address.

Without Binding

Remote $\rightarrow$ HA $\rightarrow$ Mobile

With Binding

Remote $\rightarrow$ Care-of

Cache Problem

Mobile host may move again. The cached care-of address becomes outdated. Home agent sends a warning packet.

Case Study + Key Takeaways

Professional Case Study

A mobile employee moves from the home office to a foreign network during communication.

- What remains permanent?
- Which address changes?
- Who discovers the foreign network?
- Who receives the first packet?
- Where does tunneling occur?
- How does the packet reach the mobile host?
- Why is movement transparent?

Remember

- IP originally assumes stationary hosts.
- Mobile IP uses home + care-of addresses.
- Discovery precedes registration.
- Registration precedes data transfer.
- Agents hide mobility from the Internet.
- Tunneling connects home and foreign agents.
- Double crossing and triangle routing cause inefficiency.
- Binding can reduce inefficient routing.

Complete Mobile IP flow: Addressing → Agents → Discovery → Registration → Data Transfer → Routing Efficiency